Ajay Rajendra Kumar

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Education

Northeastern University, Boston, USA — Sep. 2024 – May 2026
M.S. in Computer Science
- GPA: 4.0/4.0
- Capstone: *Dense Visual SLAM on Edge Devices*
- Advisor: Prof. Huaizu Jiang

Manipal Institute of Technology, Manipal, India — Aug. 2020 – Jul. 2024
B.Tech. in Computer Science and Engineering
- GPA: 9.3/10.0
- Minor: Computational Intelligence
- Dissertation: *Evaluating Deep Learning Approaches for Fracture Segmentation in X-ray Images*
- Advisor: Prof. Subhamoy Mandal (IIT, Kharagpur, India)

Research Experience

Culture and Cognition Lab, University of Michigan — Sep. 2026 – Present
Computational Research Specialist, working with Prof. Shinobu Kitayama — Ann Arbor, USA
- Developing agent-based simulations to investigate how intergroup threats and reputation-based cooperation shape interdependence.
- Designing controlled multimodal experiments combining long-context, multi-turn textual transcripts with facial-expression images to investigate how culturally structured visual and textual inputs shape learned representations and context-sensitive reasoning.
- Co-developing a multi-turn benchmark for evaluating culturally responsive AI social advice, defining criteria for value sensitivity, perspective-taking, and uncertainty recognition to examine their relationship with perceived helpfulness and anticipated user wellbeing; contributing to grant proposal development.

FarSight Lab, Northeastern University — Nov. 2025 – Jun. 2026
Research Assistant with Prof. Lorenzo Torresani — Boston, USA
- Improved video question answering performance of PerceptionLM-8B by 12% on the EgoBlind benchmark through prompt optimization and supervised fine-tuning; identified that abstention on safety-related questions limited the model's practical utility.
- Developed a bottom-up visual grounding pipeline that tracked visible entities and generated question-relevant textual evidence; found that even accurate evidence rarely changed incorrect model responses, revealing unreliable evidence integration.
- Improved PerceptionLM-8B performance on V-STaR by 9% and 14% in zero-shot and LoRA fine-tuned settings, respectively, using question-relevant visual guidance through SAM3-generated bounding-box overlays; investigated training a question-conditioned visual grounding module with reinforcement learning to automate visual guidance.
- Designed structured chain-of-thought supervision requiring models to identify relevant visual evidence before reasoning toward an answer; constructed 10K+ grounded training traces for parameter-efficient fine-tuning; found that models could learn to cite visual evidence without reliably grounding their reasoning in the provided regions.

Visual Intelligence Lab, Northeastern University — Apr. 2025 – Mar. 2026
Research Assistant with Prof. Huaizu Jiang — Boston, USA
- Identified fundamental limitations of sparse voxel-based visual odometry through systematic ablation studies, tracing pose estimation errors to geometric information lost at low voxel resolutions and motivating a pivot toward dense correspondence-based SLAM.
- Designed and trained a lightweight uncertainty estimation module to predict pixel-wise correspondence confidence, filtering unreliable matches to improve the robustness and accuracy of downstream camera pose estimation.
- Developed a scalable Kubric-based synthetic data generation pipeline to render 2K+ stereo video sequences with diverse camera trajectories, object configurations, and HDRI environments for training the confidence estimation network.

- Co-developed NeuSLAM, a dense visual SLAM system integrating the confidence-guided front-end with an efficient classical pose-tracking back-end, achieving 9.9% F1-all on KITTI 2015 optical flow and competitive zero-shot synthetic-to-real stereo disparity performance on KITTI 2015 and ETH3D.

D'Amore-McKim School of Business, Northeastern University Feb. 2025 – May 2026
Research Assistant with Prof. Anna Lamin and Prof. Valentina Marano Boston, USA

- Developed an end-to-end data acquisition pipeline to collect and preprocess corporate social responsibility disclosures from Indian companies, curating 100K+ records spanning five years for empirical analysis.
- Developed scalable Python-based ETL pipelines for JSON and PDF documents, incorporating structure-aware document chunking, metadata extraction, and automated preprocessing to support large-scale information retrieval and extraction.
- Developed an LLM-powered document intelligence pipeline combining semantic retrieval, targeted OCR, and OpenAI models to extract, normalize, and structure information from unstructured corporate reports for downstream empirical analysis.
- Built an embedding-based retrieval and GPT-4.1 Likert-scoring pipeline to evaluate 10K+ Crunchbase company profiles against ten research criteria, enabling analysis of the domestic market orientation of Indian firms.

Manipal Institute of Technology Jan. 2023 – Jul. 2023
Research Assistant with Prof. Nisha P. Shetty Manipal, India

- Developed and evaluated misinformation classification models in PyTorch, comparing transformer-based architectures including XLNet and BERT on the COVID-19 ANTi-Vax dataset of 15K+ tweets.
- Collected and processed community discourse from targeted vaccine-related subreddits using the Python Reddit API Wrapper (PRAW) for downstream social media analysis.
- Developed a user-association mapping algorithm based on cosine similarity and constructed interaction networks to analyze relationships among users using NetworkX.

Publications

* denotes equal contribution

International Conferences

1. Aniket Gupta, Tianye Ding, **Ajay Rajendra Kumar**, Dennis Giaya, Pia Bideau, Charles Saunders, Qu Cao, Aruni RoyChowdhury, Hanumant Singh, and Huaizu Jiang. *NeuSLAM: Dense Visual SLAM on Edge Devices*. **Spotlight** at ICRA 2026 Workshop on From Sea to Space: Advancing Perception in Harsh Domains; accepted to CVPR 2026 Demo Track. Submitted to ICRA 2027. [paper]
2. C. Rao*, G. M. Prabhu*, **Ajay Rajendra Kumar***, S. Gupta, and N. P. Shetty. *Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics*. International Conference on Machine Learning and Data Engineering (ICMLDE), 2023. [paper]

Industry Experience

Fidelity Investments Jun. 2023 – Aug. 2023
Software Development Intern Bengaluru, India

- Reduced dashboard query latency by 90% by migrating the underlying data warehouse from Oracle to Snowflake and re-architecting query execution for large-scale change-ticket analytics.
- Delivered full-stack features for an internal change-ticket management platform using Angular and Spring Boot, redesigning core workflows for creating, tracking, and managing engineering change requests.
- Designed and integrated RESTful APIs using Spring Boot to connect the Angular frontend with Snowflake-backed backend services, enabling efficient ticket creation, tracking, and reporting workflows.

Grad Course Projects

Pose Graph Sparsification via Algebraic Connectivity Mar. 2026
Mobile Robotics, Northeastern University

- Developed pose graph sparsification algorithms based on Maximizing Algebraic Connectivity (MAC) to select informative edges while reducing graph size for long-horizon SLAM.
- Implemented iterative MAC-based edge selection with continuous relaxation and Madow/nearest-neighbor rounding, using odometric backbones and minimum spanning trees to preserve graph connectivity.
- Evaluated sparsification methods across standard SLAM pose graph datasets, including KITTI, Intel, Sphere, and City datasets.

Equivariance for Text-Driven Human Motion Generation Mar. 2025
Deep Learning, Northeastern University

- Integrated equivariant neural networks into a diffusion-based Motion Diffusion Model (MDM) to enforce rotational and reflectional symmetries for text-driven human motion generation.
- Modified the MDM Transformer encoder with EquiLinear layers using scalar irreducible representations to preserve motion symmetries during generation.
- Trained and evaluated the model on HumanML3D using CLIP text embeddings, measuring generation quality and text-motion alignment with FID and R-precision.

Teaching Experience

Object-Oriented Design (CS2163) Aug. 2023 – Dec. 2023
Undergraduate Teaching Assistant, Manipal Institute of Technology Manipal, India

- Assisted 20+ undergraduate students with object-oriented programming, software design principles, and Java programming through weekly office hours and hands-on guidance.
- Supported evaluation of programming assignments and provided feedback on code quality, design, and implementation.

Problem Solving Using Computers (CS1081) Aug. 2023 – Dec. 2023
Undergraduate Teaching Assistant, Manipal Institute of Technology Manipal, India

- Guided 20+ undergraduate students in programming fundamentals, algorithmic problem solving, and debugging through weekly lab sessions and individual support.
- Evaluated programming assignments and provided feedback to help students improve their C programming skills.

Leadership

Indian Society for Technical Education (ISTE), Manipal Chapter Jun. 2022 – Dec. 2023
Machine Learning Head, Manipal Institute of Technology Manipal, India

- Led the chapter's machine learning initiatives, coordinating a team of 30+ students and organizing technical workshops, seminars, and hands-on sessions.
- Mentored 3 student teams on machine learning projects, providing guidance on problem formulation, model development, experimentation, and evaluation.
- Collaborated with student organizers and faculty members to develop the chapter's technical programs and expand student engagement in machine learning.

Open Source

TorchMetrics, Contributor 2026

- Fixed a Transformers compatibility issue in CLIPScore by handling dataclass-based feature outputs and adding regression tests; merged upstream (#3408).

Technical Skills

Programming: Python, C/C++, Java, SQL, R

Frameworks & Tools: PyTorch, TensorFlow, vLLM, Hugging Face, PEFT (LoRA, QLoRA), bitsandbytes, Weights & Biases, Git

Distributed Training: PyTorch Distributed (DDP/FSDP), Slurm, DeepSpeed

Web & Data: Angular, Spring Boot, Snowflake, Oracle DB

Honors & Awards

Mar. 2025	Most Innovative Project Award, Deep Learning (CS7150)	Boston, USA
Oct. 2023	Undergraduate Research Sponsorship Grant	Dehradun, India
Aug. 2022	National 2nd Runner-Up, Daimler India Hackathon	Chennai, India
Apr. 2022	1st Runner-Up, Transfusion Medicine Hackathon	Manipal, India
Jun. 2021	Winner, Space Odyssey Web Development Hackathon	Manipal, India
Jan. 2021	Achiever's Scholarship Award	Manipal, India

Community Involvement

Child Rights and You (CRY), *Outreach Volunteer*	Mar. 2023 – Aug. 2024
AIESEC, *Outgoing Global Talent Department, Mentor*	Mar. 2023 – Sep. 2023
Indian Association of Physiotherapists, *COVID-19 Chatbot Developer*	May 2021 – Jun. 2021